Proposta de tema e plano de trabalhos para projeto de Computação/Tecnologias Quânticas

Título: Image classification with Hybrid Quantum Convolutional Neural Networks

Nº alunos: 1, 2, ...

Orientador/Orientadores : Jorge Lobo, Gabriel Falcão

Enquadramento: Computação Quântica/ ~~Tecnologias Quânticas~~

Descrição e objetivos:

Artificial neural networks are currently one of the most popular models used in machine learning. A subcategory of these called convolutional neural networks (CNN) is particularly popular. These are networks mainly used for the task of image recognition. Using pooling and convolution layers they extract structured information about images. My project will be related to CNN topic but also to Quantum computing. Quantum machine learning leverages the principles of quantum mechanics to process information, allowing for the creation of algorithms that can potentially outperform classical counterparts. By harnessing quantum bits or qubits, these algorithms can explore multiple possibilities simultaneously, offering a unique approach to solving complex problems. While still in the early stages of development, quantum machine learning holds promise for revolutionizing certain computational tasks, particularly those involving large-scale optimization and data analysis.

Plano de trabalhos:

This project will involve using the Qiskit framework combined with Pytorch to construct a HQCNN performing the task of image classification. The project described in the Qiskit tutorial [4] can be used as a starting point. This project can be divided into four steps:

Referências

[1] Iris Cong, Soonwon Choi and Mikhail D. Lukin1, "Quantum convolutional neural networks" Nature physics 2019.

[2] Naim Ajlouni, Adem O¨zyavas¸, Mustafa Takaog˘lu, Faruk Takaog˘lu and Firas Ajlouni, "Medical image diagnosis based on adaptive Hybrid Quantum CNN" BMC Medical Imaging, 2023.

[3] Chen, G., Chen, Q., Long, S. et al. Quantum convolutional neural network for image classification. Pattern Anal Applic 26, 655–667 (2023). https://doi.org/10.1007/s10044-022-01113-z

[4]https://qiskit-community.github.io/qiskit-machine-learning/tutorials/05_torch_connector.html

[5] http:///pytorch.org/docs/stable/index.html